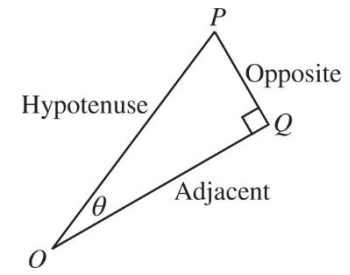
Evaluate Trigonometric Functions of Any Angle

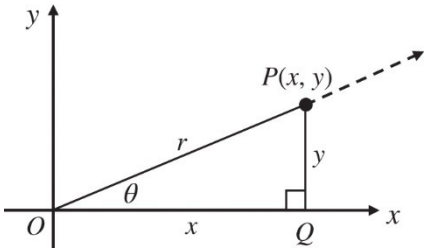
ΔPOQ is a right triangle with acute angle θ.



We can orient the triangle with angle θ in standard position, and label a point $P(x, y)$ on the terminal side of θ.

The length of the horizontal leg is x and the length of the vertical leg is y.

The hypotenuse $r = \sqrt{x^2 + y^2}$ is the distance between $P(x, y)$ and $(0, 0)$.

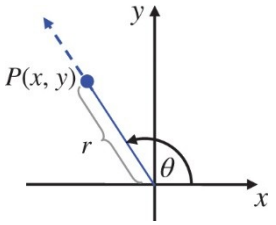


Relative to angle θ, side y is the opposite side, x is the adjacent side, and r is the hypotenuse.

From the right triangle definition of the six trigonometric functions, we have:

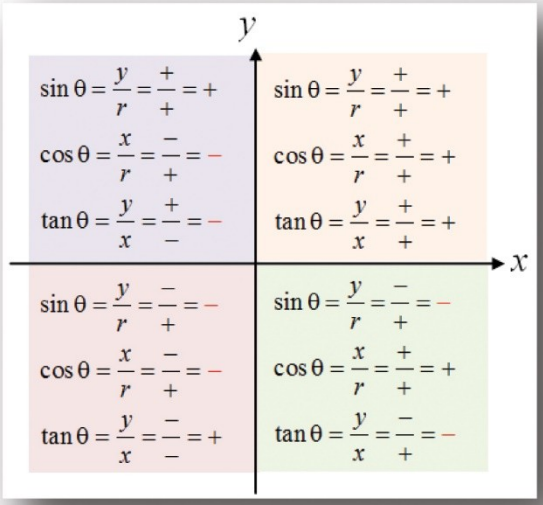
$$\sin\theta = \frac{\text{opp}}{\text{hyp}} = \frac{y}{r} \qquad \cos\theta = \frac{\text{adj}}{\text{hyp}} = \frac{x}{r} \qquad \tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{y}{x}$$

$$\csc\theta = \frac{\text{hyp}}{\text{opp}} = \frac{r}{y} \qquad \sec\theta = \frac{\text{hyp}}{\text{adj}} = \frac{r}{x} \qquad \cot\theta = \frac{\text{adj}}{\text{opp}} = \frac{x}{y}$$



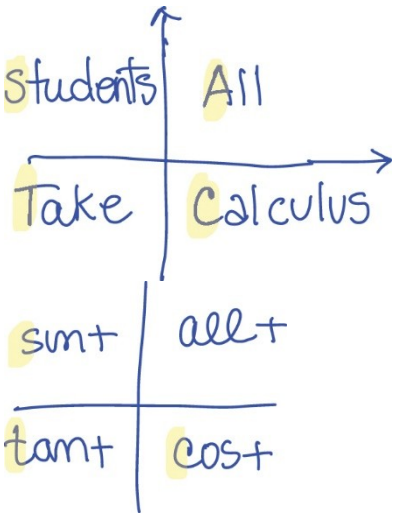
These definitions may be extended to angles that are not acute. (Assume none of the denominators are zero.)

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The signs of the values of the trigonometric functions depend solely on the signs of x and y.

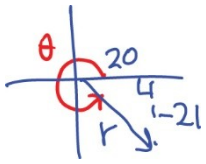
The reciprocal functions cosecant, secant, and cotangent, respectively will have the same signs as sine, cosine, and tangent.



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$P(20,-21)$

1. Let be a point on the terminal side of angle θ drawn in standard position. Find the value of $\sin\theta$ and $\cot\theta$.



$$\sin\theta = \frac{y}{r} = \frac{-21}{29} = -\frac{21}{29}$$

$$\cot\theta = \frac{x}{y} = \frac{20}{-21} = -\frac{20}{21}$$

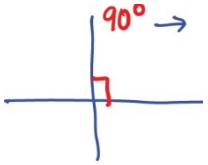
$x = 20,\ y = -21$

$r = \sqrt{20^2 + (-21)^2}$

$= 29$

Angles with terminal sides on the coordinate axes are termed quadrantal angles. In the case of these angles, ratios with $x = 0$ or $y = 0$ in the denominator are undefined.

2. Find the sine, cosine, and tangent for the angle $\theta = 90°$.



$x = 0$

$y = 1$

$r = \sqrt{0^2 + 1^2} = 1$

$$\sin 90° = \frac{y}{r} = \frac{1}{1} = 1$$

$$\cos 90° = \frac{x}{r} = \frac{0}{1} = 0$$

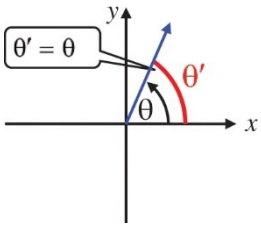
$$\tan 90° = \frac{y}{x} \to \frac{1}{0}$$

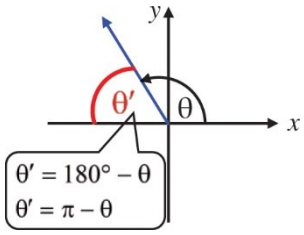
undefined

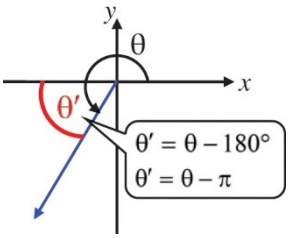
Determine Reference Angles

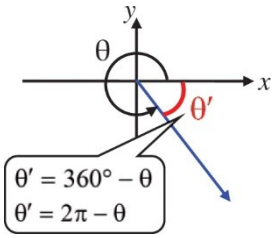
θ'

Reference Angle: Let θ be an angle in standard position. The **reference angle** for θ is the acute angle formed by the terminal side of θ and the horizontal axis.





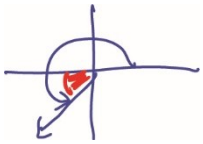




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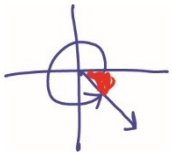
3. Find the reference angle for the given angle.

a. $225°$



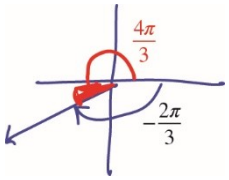
$225° - 180° = \boxed{45°}$

b. $\dfrac{11\pi}{6}$



$2\pi - \dfrac{11\pi}{6} = \boxed{\dfrac{\pi}{6}}$

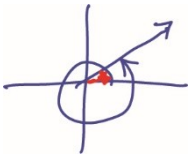
c. $-\dfrac{2\pi}{3}$



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$$\frac{4\pi}{3} - \pi = \boxed{\frac{\pi}{3}}$$

d. $390°$



$$390° - 360° = 30°$$

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Evaluate Trigonometric Functions Using Reference Angles

To find the value of a trigonometric function of a given angle θ,

1. Determine the function value of the reference angle θ'.
2. Affix the appropriate sign based on the quadrant in which θ lies.

Use reference angles to find the exact value.

4. $\cos 135° = -\cos 45° = \dfrac{-\sqrt{2}}{2}$



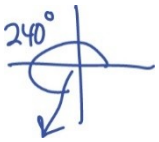
Q II cosine negative

$180° - 135° = 45°$

$\cos 45° = \dfrac{\sqrt{2}}{2}$

5. $\sin 240° = -\sin 60° = \dfrac{-\sqrt{3}}{2}$

Q III sine negative



$240° - 180° = 60°$

$\sin 60° = \dfrac{\sqrt{3}}{2}$

6. $\tan\dfrac{5\pi}{6} = -\tan\dfrac{\pi}{6} = -\dfrac{\sqrt{3}}{3}$

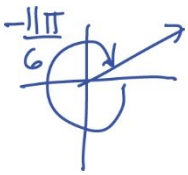


Q II tangent negative

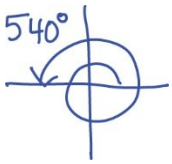
$\pi - \dfrac{5\pi}{6} = \dfrac{\pi}{6}$

$\tan\dfrac{\pi}{6} = \dfrac{\sqrt{3}}{3}$

7. $\csc\left(-\dfrac{11\pi}{6}\right) = \csc\dfrac{\pi}{6} = 2$



8. $\cot 540° = \cot 180°$



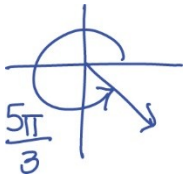
undefined

$540° - 360° = 180°$

$\cot 180°$ undefined

9. $\sec\dfrac{5\pi}{3} = \sec\dfrac{\pi}{3} = 2$

Q IV secant positive



$2\pi - \dfrac{5\pi}{3} = \dfrac{\pi}{3}$

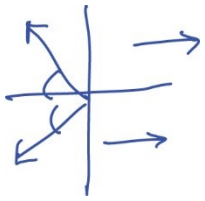
$\sec\dfrac{\pi}{3} = 2$

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10. Find two angles between 0 and 360° where $\cos\theta = -\frac{\sqrt{3}}{2}$.

$\cos 30° = \frac{\sqrt{3}}{2}$

cosine neg in QII, III



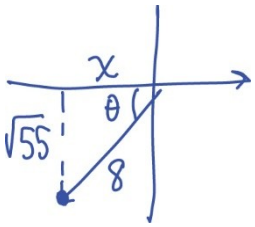
$180° - 30° = 150°$

$180° + 30° = 210°$

$\theta = 150°, 210°$

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11. Given $\sin\theta = -\frac{\sqrt{55}}{8}$ and $\cos\theta < 0$, find $\cot\theta$ and $\cos\theta$.



Sine neg, cosine neg in Q III

$$x^2 + \left(\sqrt{55}\right)^2 = 8^2$$
$$x^2 + 55 = 64$$
$$x^2 = 9$$
$$x = 3$$

$x = -3$, $y = -\sqrt{55}$, $r = 8$

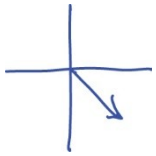
$$\cos\theta = -\frac{3}{8}$$
$$\cot\theta = \frac{3}{\sqrt{55}}\cdot\frac{\sqrt{55}}{\sqrt{55}} = \frac{3\sqrt{55}}{55}$$

Pythagorean Identities	Reciprocal Identities	Quotient Identities
$\sin^2\theta + \cos^2\theta = 1$	$\csc\theta = \frac{1}{\sin\theta}$	$\tan\theta = \frac{\sin\theta}{\cos\theta}$
$\tan^2\theta + 1 = \sec^2\theta$	$\sec\theta = \frac{1}{\cos\theta}$	$\cot\theta = \frac{\cos\theta}{\sin\theta}$
$1 + \cot^2\theta = \csc^2\theta$	$\cot\theta = \frac{1}{\tan\theta}$	

---

12. Given $\sec\theta = \frac{41}{40}$ for θ in Quadrant IV, use fundamental identities to find tan θ and sin θ.

$$\tan^2\theta + 1 = \sec^2\theta$$
$$\tan^2\theta + 1 = \left(\frac{41}{40}\right)^2$$
$$\tan^2\theta = \frac{1681}{1600} - 1$$
$$\tan^2\theta = \frac{81}{1600}$$
$$\tan\theta = \pm\sqrt{\frac{81}{1600}} = \pm\frac{9}{40}$$



Q IV tangent neg

$$\tan\theta = -\frac{9}{40}$$

$$\sec\theta = \frac{41}{40} \qquad \sec\theta = \frac{1}{\cos\theta}$$
$$\frac{1}{\cos\theta} = \frac{41}{40} \rightarrow \cos\theta = \frac{40}{41}$$
$$\sin^2\theta + \cos^2\theta = 1$$
$$\sin^2\theta + \left(\frac{40}{41}\right)^2 = 1$$
$$\sin^2\theta = 1 - \frac{1600}{1681}$$
$$\sin^2\theta = \frac{81}{1681}$$
$$\sin\theta = \pm\sqrt{\frac{81}{1681}} = \pm\frac{9}{41}$$

QIV sine neg

$$\sin\theta = -\frac{9}{41}$$

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